

Whole Exome Sequencing Analysis

Patient name: XX	PIN: XX
Age: 26 Years	Gender: Male
Hospital/Clinic: XX	Sample Number: XX
Referring Clinician: XX	Sample Collection Date: NA
Specimen: Peripheral Blood	Sample Received Date: NA
	Report Date: NA

Clinical History

Proband, XX is born of a non-consanguineous marriage. XX is presented with chronic kidney disease (on continuous ambulatory peritoneal dialysis (CAPD) since 2022 and hypertension. On examination, hemoglobin of 9.6 g/dL, serum creatinine of 16.01 mg/dL. There is a family history of mother (deceased) affected with kidney disease. Proband, XX is suspected to be affected with hereditary glomerulonephritis and has been evaluated for pathogenic variations.

Results

Variant of uncertain significance causative to the related phenotype was identified

List of significant variant(s) identified:

Gene	Variant*	Allele Status	Region	Disease or Disorder	Inheritance	Classification*
WT1 (-) (NM_024426.6)	c.1088C>T (p.Thr363Met)	Heterozygous	Exon 6	Meacham syndrome (OMIM# 608978) Nephrotic syndrome, type 4 (OMIM# 256370)	Autosomal Dominant	Uncertain Significance (PM2_Supp, PP3)

Variant interpretation:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
WT1 (-)	c.1088C>T (p.Thr363Met)	Autosomal Dominant	Meacham syndrome (OMIM#608978) Nephrotic syndrome, type 4 (OMIM#256370)	PM2_Supp, PP3	Uncertain Significance
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 6	Heterozygous	Missense variant	Not observed in annotated gnomAD v4 All	Ref(G): 70, Alt(A): 64, VAF: 47.76%	
In silico Prediction Tools				Conservation Status	
SIFT: Da	LRT: U	PolyPhen2: Da	CADD: U	MT2: D	PhyloP: C GERP++: C
Genomic Position			Clinical Evidence		
Chr11:NC_000011.10:g.32399973G>A (NM_024426.6)			ClinVar: 2935357		

Abbreviation: **D:** Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain; **Supp:** Supporting;

Clinical Evidence: This variant has been classified as uncertain significance in ClinVar database.

*Genetic test results are based on the recommendation of American college of Medical Genetics [1]. No other variant that warrants to be reported for the given clinical indication was identified

CNV Findings

No significant Copy Number Variations (CNV) related to phenotype was detected.

Recommendations

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the blood was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using GenolabM sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS- SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported. Gene list used in the assay [Click here](#).

Sequence data attributes

Total Read Generated	11.90 Gb
Data ≥ Q30	94.31%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions. The variants of uncertain significance and variations with high minor allele frequencies which are likely to be benign will be given upon request.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA*, and *CYP21A2* which have very high homology. Moreover, the *ORF15* transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

Reference

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in medicine: official journal of the American College of Medical Genetics vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

This report has been reviewed and approved by:

				
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Appendix 1: Gene Coverage Indication Based Analysis:

Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered
ABCC6	100.0	ABCD4	100.0	ACE	100.0	ACP5	100.0
ACTA2	100.0	ACTB	100.0	ACTG2	100.0	ACTN4	100.0
ACVRL1	100.0	ADAMTS13	100.0	AFF4	100.0	AGPAT2	100.0
AGT	100.0	AGTR1	100.0	AGXT	100.0	AHL1	100.0
ALG1	100.0	ALG8	100.0	ALG9	100.0	ALMS1	100.0
ALPL	100.0	AMER1	100.0	AMN	100.0	ANKS6	100.0
ANLN	100.0	ANOS1	100.0	AP2S1	100.0	APC2	100.0
APOA1	100.0	APOE	100.0	APOL1	100.0	APRT	100.0
AQP2	100.0	ARHGAP24	100.0	ARHGDIA	100.0	ARID1A	100.0
ARID1B	100.0	ARL13B	99.97	ARL6	100.0	ARMCS	100.0
ARNT2	100.0	ARX	100.0	ASXL1	100.0	ATP6V0A4	100.0
ATP6V1B1	100.0	ATP7A	100.0	ATP7B	100.0	AUH	100.0
AVIL	100.0	AVP	100.0	AVPR2	100.0	B2M	100.0
B3GLCT	100.0	B4GAT1	100.0	B9D1	100.0	B9D2	100.0
BBIP1	100.0	BBS1	100.0	BBS10	100.0	BBS12	100.0
BBS2	100.0	BBS4	100.0	BBS5	100.0	BBS7	100.0
BBS9	100.0	BCOR	100.0	BCS1L	100.0	BMP4	100.0
BMPER	100.0	BRIP1	100.0	BSCL2	100.0	BSND	100.0
BUB1B	100.0	C1QA	100.0	C1QB	100.0	C1QC	100.0
C2	100.0	C3	100.0	CA2	100.0	CACNA1S	100.0
CAD	100.0	CASP10	100.0	CASR	100.0	CC2D2A	100.0
CCBE1	100.0	CCDC22	100.0	CD151	100.0	CD19	100.0
CD2AP	100.0	CD81	100.0	CD96	100.0	CDC5L	100.0
CDC73	100.0	CDKN1B	100.0	CDKN1C	100.0	CENPF	100.0
CEP104	100.0	CEP120	100.0	CEP164	100.0	CEP290	100.0
CEP41	100.0	CEP83	100.0	CFH	100.0	CFHR5	100.0
CFI	100.0	CHD7	100.0	CHRM3	100.0	CHST14	100.0
CISD2	100.0	CLCN5	100.0	CLCNKA	100.0	CLCNKB	100.0
CLDN16	100.0	CLDN19	100.0	CNNM2	100.0	COL18A1	100.0
COL4A1	100.0	COL4A3	100.0	COL4A4	100.0	COL4A5	100.0
COL5A1	100.0	COPA	100.0	COQ2	100.0	COQ6	100.0
COQ7	100.0	COQ8B	100.0	COQ9	100.0	COX10	100.0
COX14	100.0	COX20	100.0	COX6B1	100.0	COX7B	100.0
COX8A	100.0	CPT1A	100.0	CPT2	100.0	CRB2	100.0
CREBBP	100.0	CRTAP	100.0	CSPP1	100.0	CTC1	100.0
CTNS	100.0	CUBN	100.0	CUL3	100.0	CYP11A1	100.0
CYP11B1	100.0	CYP17A1	100.0	CYP21A2	100.0	CYP24A1	100.0
DCDC2	100.0	DCHS1	100.0	DDX59	100.0	DGKE	100.0
DHCR7	100.0	DIS3L2	100.0	DKC1	100.0	DLG3	100.0
DLL3	100.0	DLL4	100.0	DLX4	100.0	DMP1	100.0
DNA2	100.0	DNAAF1	100.0	DNAJB11	100.0	DNAJB9	100.0
DNASE1L3	100.0	DNMT3B	100.0	DPH1	100.0	DSTYK	100.0
DYNC2H1	100.0	EBP	100.0	EDNRA	100.0	EFEMP2	100.0
EGF	100.0	EHHADH	100.0	EIF2AK3	100.0	EIF2B4	100.0
EMP2	100.0	ENG	100.0	ENPP1	100.0	ERCC6	100.0
EPG5	100.0	ERBB3	100.0	ERCC4	100.0	ETFB	100.0
ERCC8	100.0	ESCO2	99.95	ETFA	100.0	EYA1	100.0
ETFDH	100.0	EVC	100.0	EVC2	100.0	FAN1	100.0
FAH	100.0	FAM20A	100.0	FAM20C	100.0	FANCD2	100.0
FANCA	100.0	FANCB	100.0	FANCC	100.0	FANCI	100.0
FANCE	100.0	FANCF	100.0	FANCG	100.0	ZNF687	100.0

FANCL	100.0	FANCM	100.0	FASTKD2	100.0	FAT1	100.0
FAT4	100.0	FBLN5	100.0	FBXL4	100.0	FGA	100.0
FGF10	100.0	FGF20	100.0	FGF23	100.0	FGFR1	100.0
FGFR2	100.0	FGFR3	100.0	FKBP14	100.0	FLCN	100.0
FLNA	100.0	FLNB	100.0	FLT4	100.0	FN1	100.0
FOXC1	100.0	FOXC2	100.0	FOXF1	100.0	FRAS1	100.0
FREM1	100.0	FREM2	100.0	FUZ	100.0	FXYD2	100.0
GALNT3	100.0	GATA3	100.0	GATA6	100.0	GBA	99.81
GCDH	100.0	GCM2	100.0	GDF2	100.0	GDNF	100.0
GLA	100.0	GLB1	100.0	GLI3	100.0	GLIS2	100.0
GLIS3	100.0	GNA11	100.0	GNAS	100.0	GNB1	100.0
GPC3	100.0	GPC5	100.0	GRHPR	100.0	GRIP1	100.0
GSN	100.0	HBB	100.0	HDAC8	100.0	HES7	100.0
HGD	100.0	HNF1A	100.0	HNF1B	100.0	HNF4A	100.0
HOGA1	100.0	HOXA13	100.0	HOXD13	100.0	HPRT1	100.0
HPS1	100.0	HPSE2	100.0	HRAS	100.0	HSD11B2	100.0
HSD17B3	100.0	HSD17B4	100.0	HSD3B2	100.0	HSPA9	100.0
HSPG2	100.0	HYLS1	100.0	IFT122	100.0	IFT140	100.0
IFT172	100.0	IFT27	100.0	IFT43	100.0	IFT80	100.0
INF2	98.87	INPP5E	100.0	INPL1	100.0	INSR	100.0
INVS	100.0	IQCB1	100.0	IRF6	100.0	ITGA3	100.0
ITGA6	100.0	ITGA8	100.0	ITGB4	100.0	JAG1	100.0
JAM3	100.0	KANK1	99.43	KANK2	100.0	KANK4	100.0
KANSL1	100.0	KAT6B	100.0	KCNA1	100.0	KCNH1	100.0
KCNJ1	100.0	KCNJ10	100.0	KCNJ5	100.0	KCTD1	100.0
KDM1A	100.0	KDM6A	100.0	KIF14	100.0	KIF7	100.0
KL	100.0	KLHL3	100.0	KMT2D	100.0	KRAS	100.0
KYNU	100.0	LAGE3	100.0	LAMB2	100.0	LAMB3	100.0
LAMC2	100.0	LCAT	100.0	LDHA	100.0	LFNG	100.0
LMBRD1	100.0	LMNA	100.0	LMX1B	100.0	LONP1	100.0
LPIN1	100.0	LRIG2	100.0	LRP2	100.0	LRP4	100.0
LTBP4	100.0	LYZ	100.0	LZTFL1	100.0	MAFB	100.0
MAGED2	100.0	MAGI2	100.0	MAP2K1	100.0	MAP2K2	100.0
MAPRE2	100.0	MBTPS2	100.0	MEFV	100.0	MESP2	100.0
MKKS	100.0	MKS1	100.0	MLH1	100.0	MMACHC	100.0
MXN1	100.0	MRPS22	100.0	MSH2	100.0	MSH6	100.0
MTM1	100.0	MVK	100.0	MYCN	100.0	MYH9	100.0
MYO1E	100.0	NAA10	100.0	NARS2	100.0	NBN	100.0
NDUFB2	100.0	NECTIN1	100.0	NEK1	100.0	NEK8	100.0
NEU1	100.0	NEXMIF	100.0	NF1	100.0	NHP2	100.0
NIPBL	100.0	NLRP3	100.0	NOTCH2	99.41	NOTCH3	100.0
NPHP1	100.0	NPHP3	100.0	NPHP4	100.0	NPHS1	100.0
NPHS2	100.0	NR0B1	100.0	NR3C2	100.0	NSD1	100.0
NSDHL	100.0	NUP107	100.0	NUP133	100.0	NUP205	100.0
NUP85	100.0	NUP93	100.0	NXF5	100.0	OCLN	97.33
OCRL	100.0	OFD1	100.0	OPLAH	100.0	OSGEP	100.0
PAF1	100.0	PALB2	100.0	PAX2	100.0	PC	100.0
PDE6D	100.0	PDSS1	100.0	PDSS2	100.0	PET100	100.0
PEX1	100.0	PEX10	100.0	PEX11B	100.0	PEX12	100.0
PEX13	100.0	PEX14	100.0	PEX16	100.0	PEX19	100.0
PEX2	100.0	PEX26	100.0	PEX3	100.0	PEX5	100.0
PEX6	100.0	PGK1	100.0	PGM3	100.0	PHEX	100.0
PHGDH	100.0	PIEZO2	100.0	PIGA	100.0	PIGL	100.0
PIGN	100.0	PIGT	100.0	PIK3CA	100.0	PIK3R2	100.0
PITX2	100.0	PKD1	100.0	PKD2	100.0	PKHD1	100.0
PLCE1	100.0	PLG	100.0	PLOD1	100.0	PMM2	100.0
PNPLA6	100.0	PODXL	100.0	POMC	100.0	POMT1	100.0
POR	100.0	PORCN	100.0	PPP1R15B	100.0	PQBP1	100.0

PRKCD	100.0	PROKR2	100.0	PRPS1	100.0	PSAP	100.0
PTEN	100.0	PTH	100.0	PTH1R	100.0	PTPN11	100.0
PTPRO	100.0	PUF60	100.0	PYGM	100.0	RAB18	100.0
RAB23	100.0	RAB3GAP1	100.0	RAB3GAP2	100.0	RAD51C	100.0
RAI1	100.0	RAP1A	100.0	RAP1B	100.0	RBBP8	100.0
RBM10	100.0	RBM8A	100.0	RECQL4	100.0	REN	100.0
REER	100.0	RET	100.0	RIN2	100.0	RIPK4	100.0
RIPPLY2	100.0	RMND1	100.0	ROBO2	100.0	ROR2	100.0
RPGRIPI1L	100.0	RPL11	100.0	RPL26	100.0	RPL5	100.0
RPS10	100.0	RPS17	100.0	RPS19	100.0	RPS24	100.0
RPS26	100.0	RPS28	100.0	RPS29	100.0	RPS7	100.0
RRM2B	100.0	RITN	100.0	SALL1	100.0	SALL4	100.0
SARS2	100.0	SBD5	100.0	SC5D	100.0	SCARB2	100.0
SCN4A	100.0	SCNN1A	100.0	SCNN1B	100.0	SCNN1G	100.0
SCO1	100.0	SDCCAG8	100.0	SDHB	100.0	SDHC	100.0
SDHD	100.0	SEMA3E	100.0	SERPINH1	100.0	SETBP1	100.0
SF3B4	100.0	SGPL1	100.0	SHH	100.0	SI	100.0
SIX1	100.0	SIX2	100.0	SIX5	100.0	SLC12A1	100.0
SLC12A3	100.0	SLC16A12	100.0	SLC1A1	100.0	SLC22A12	100.0
SLC25A1	100.0	SLC26A4	100.0	SLC2A10	100.0	SLC2A2	100.0
SLC2A9	100.0	SLC34A1	100.0	SLC34A3	100.0	SLC36A2	100.0
SLC37A4	100.0	SLC3A1	100.0	SLC4A1	100.0	SLC4A4	100.0
SLC5A2	100.0	SLC6A19	100.0	SLC6A20	100.0	SLC7A7	100.0
SLC7A9	100.0	SLC9A3R1	100.0	SLIT2	100.0	SLX4	100.0
SMAD3	100.0	SMARCA1	100.0	SMARCE1	100.0	SMC1A	100.0
SMOC1	100.0	SNRPB	100.0	SOX11	100.0	SOX17	100.0
SOX18	100.0	SOX9	100.0	SPECC1L	100.0	SPINT2	100.0
SRCAP	100.0	STAR	100.0	STK11	100.0	STRA6	100.0
STRADA	100.0	STUB1	100.0	STX16	100.0	SUCLA2	100.0
SUGCT	100.0	SYNPO	100.0	TACO1	100.0	TAPT1	100.0
TBC1D20	100.0	TBC1D24	100.0	TBCE	100.0	TBX18	100.0
TCTN2	100.0	TCTN3	100.0	TERC	100.0	TFAP2A	100.0
THOC6	100.0	TMCO1	100.0	TMEM138	100.0	TMEM216	100.0
TMEM231	100.0	TMEM237	100.0	TMEM67	100.0	TMEM70	100.0
TNFRSF1A	100.0	TNXB	97.75	TP53RK	100.0	TP63	100.0
TPRKB	100.0	TRAF3IP1	100.0	TRAIP	100.0	TRAP1	100.0
TRMT5	100.0	TRNT1	100.0	TRPC6	100.0	TRPM6	100.0
TSC1	100.0	TSC2	100.0	TSR2	100.0	TTC21B	100.0
TTC37	100.0	TTC8	100.0	TTR	100.0	TWIST2	100.0
TXNL4A	100.0	UBE2T	100.0	UBR1	100.0	UMOD	100.0
UMPS	100.0	UPB1	100.0	UPK3A	100.0	UQCC2	100.0
USP9X	100.0	VANGL1	100.0	VHL	100.0	VIPAS39	100.0
WAS	100.0	WDPCP	100.0	WDR19	100.0	WDR35	100.0
WDR73	100.0	WFS1	100.0	WINK1	100.0	WINK4	100.0
WNT3	100.0	WNT4	100.0	WNT5A	99.83	WNT7A	100.0
WT1	100.0	XDH	100.0	XPNPEP3	100.0	XRCC4	100.0
XYLT2	100.0	YAP1	100.0	ZAP70	100.0	ZIC3	100.0
ZMPSTE24	100.0	ZNF423	100.0				